

Holly Huey

hhuey@adobe.com

hollyhuey.github.io | [linkedin](#) | [google scholar](#)

Building tools to enhance human creativity. I specialize in human & auto eval of GenAI media models (image/video/vector). My research statistically compares human vs. model behavior to improve creative technologies. I build auto evaluators, model orchestrators, and agents related to: *text-to-visual* & *visual-to-visual* systems, *usability*, *editability*

Experience

Applied Scientist | Technical Lead for Firefly Image Eval | Adobe

Jan 2025 – present, San Francisco, CA

- Evaluate GenAI media models (image/vector/video) against curated eval data to measure model improvement & regression before public release
- Lead human & auto evals for Firefly, Photoshop, Illustrator, and Adobe Research to fine-tune models for market & custom enterprise clients
- Build teams of professional photographers & editors, graphic designers, and video creators/editors to design custom assets and score models
- Collaborate closely with AI Safety & IP Guardrails teams

Graduate Research Scientist (Founding Lab Member) | UC San Diego

Sept 2019 – Aug 2024, San Diego, CA

- Specialized in creating new human eval tools for E2E research involving large-scale crowdsourcing of media (e.g., >90K human & GenAI sketches) & computer vision techniques to analyze visual content creation
- Performed statistical analyses (mixed-effects models, GLM, ANOVA) & generated data visualizations (ggplot, seaborn) for publications & talks
- Collaborated with cross-functional teams spanning multiple universities to investigate object recognition & abstraction across large-scale datasets
- Taught UG classes of 300+ students covering UXR, stats, cognitive psychology, and child development
- Visiting Researcher at Stanford (Jan – March 2024)

Research Scientist/Engineer Intern | Adobe Research

May – Aug 2024, New York City, NY

- Led user research developing system parameters for video editing style transfer LLM models by interviewing video creators, editors & directors

March – Aug 2024, remote

- Evaluated suite of text-to-image models on fine-grain prompt coherence related to understanding of e.g., number, physics, & spatial reasoning

June – Dec 2023, San Francisco, CA

- Led user research evaluating how video creators' different workflows and goals impact their text-to-image preferences, relative to LLM predictions

Founding Lead Researcher & Lab Manager | NYU

July 2017 – Aug 2019, New York City, NY

- Led research and designed 3D animations for studies investigating navigation, object recognition & symbolic reasoning in children & adults
- Mentored 8 honors thesis/grant-sponsored students & trained teams of 10-15 researchers (Created interdepartmental workshop series aimed at increasing UG computational literacy & research skills by teaching R, PsychoPy, Adobe CC, Blender)

Research Assistant | MIT & Harvard

Oct 2016 – June 2017, Boston, MA

- Conducted eye-tracking studies investigating collision event inferences
- Conducted pro-social studies investigating children's inferences about other agents' mental models from agent behaviors

Education

UC San Diego

PhD in Experimental Psychology | 2024

MA in Experimental Psychology | 2021

St. John's College

BA in Liberal Arts | 2016

Dual major: History of Math & Sciences, Philosophy

Dual minor: Comparative Literature, Classics

Selected GenAI Media Expertise

Human & Auto Evals

- human annotation • behavioral benchmarking
- auto evals & validation • model orchestrators
- creative agents • large-scale statistical methods

Training Data & Curated Eval Sets

- data curation & labeling • specialist team hiring
- image, vector, video, and audio asset curation
- cross-cultural & non-English benchmarks

Model Quality

- PRD co-creation on user alignment on quality, latency, cost, seed consistency, & customization
- market vs. enterprise user product direction

Leadership

- cross-functional product design & eval direction
- hiring lead for global artist teams for evals
- academic team lead & founding member
- GenAI startup & VC experience
- research talks • scientific writing

Selected Academic Publications

Semantic Segmentation in Sketches

Long, Fan, **Huey**, Chai, & Frank. (2024). Parallel developmental changes in children's production and recognition of line drawings of visual concepts. *Nature Communications*. [link](#)

Mukherjee*, **Huey***, Vinker, Aguina-Kang, Shamir, & Fan. (2023). SEVA: Leveraging sketches to evaluate alignment between human and machine visual abstraction. *NeurIPS Datasets & Benchmarks*. [link](#)

Huey, Lu, Walker, & Fan. (2022). Explanatory drawings prioritize functional properties at the expense of visual fidelity. *Cognition*. [link](#)

Data Visualization Design

Brockbank, Verma, Lloyd, **Huey**, Padilla, & Fan. (2025). *Cognitive Research: Principles & Implications*. [link](#)

Video Editing & Production

Huey, Leake, Aneja, Fisher, & Fan. (2024). How do video content creation goals impact which concepts people prioritize for generating B-roll imagery? *Creativity & Cognition*. [link](#)